Week of 31 Oct – 6 Nov 2026

6 top picks · 0 new tools · 1 worth a look

Glossary of terms used this week

Drift metric	A measured distance between synthetic and real data. Higher drift predicts a worse drop in accuracy when a model trained on synthetic data meets real records.
Jensen–Shannon divergence	A symmetric measure of how different two probability distributions are. Zero means identical; the study this week found it the best single predictor of transfer loss.
Verified self-training	Generate synthetic data, filter it with a verifier score, retrain on what survives. Repeating this on a model's own output is what causes model collapse.
Visit sequence	In sequential synthesis, the order in which variables are generated. Each variable is conditioned on the ones before it, so the order changes the quality of the result.
Fully conditional specification	The alternative to sequential synthesis: generate each variable from all the others rather than from a fixed order. More flexible, more hyperparameters, more compute.
Differentially private synthetic data	Synthetic records produced with a mathematical guarantee that no individual in the source data can be identified from the output. The privacy budget ε sets the trade-off: a lower ε means stronger privacy but usually lower analytical utility.
Synthetic control arm	A simulated control group for a clinical trial, generated from prior trials and real-world data instead of recruiting and randomising real patients.

Trend of the week

When sites train generators on uneven shards, the synthetic union can keep the counts and still miss the joints. A small experiment on how hospitals can train a shared medical AI model without sharing patient data. I'm a software engineer, building Misata Studio.

Top picks

1 nathanielkang/CrossSynth (stars: 0)
GitHub · 31 Oct 2026

When sites train generators on uneven shards, the synthetic union can keep the counts and still miss the joints. CrossSynth is a federated tabular synthesizer. Each party trains a local diffusion model under differential privacy.

2 Sejal116/federated-hospitals-simulation (stars: 0)
GitHub · 4 Nov 2026

A small experiment on how hospitals can train a shared medical AI model without sharing patient data. It compares training at one hospital alone, training on pooled data, and federated learning, then tests two privacy questions: what privacy noise costs in accuracy, and whether an attacker can tell who was in the training data. This project follows my published literature review: Gupta, S.

3 Pepperwind, a fictional company dataset with 223 tables and 79M rows
Hacker News · 2 Nov 2026

I'm a software engineer, building Misata Studio. (full text unavailable)

4 opendp/opendp (stars: 447)
GitHub · 4 Nov 2026

The OpenDP Library is a modular collection of statistical algorithms that adhere to the definition of differential privacy. It can be used to build applications of privacy-preserving computations, using a number of different models of privacy. OpenDP is implemented in Rust, with bindings for easy use from Python and R.

5 Structured Priors: Towards an algebra of component priors for prior-data fitted networks
arXiv · 2 Nov 2026

Computer Science > Machine Learning [Submitted on 2 Nov 2026] Title:Structured Priors: Towards an algebra of component priors for prior-data fitted networks View PDF HTML (experimental) Abstract:Prior-data fitted networks are typically trained using synthetic data generated by a single, manually constructed generator.

6 Microsoft Copilot Synthetic Data Targets Enterprise Evaluation's Privacy Gap - remio
remio.ai · 31 Oct 2026

Microsoft Copilot Synthetic Data Targets Enterprise Evaluation's Privacy Gap Microsoft disclosed an enterprise evaluation platform on October 30, 2026, testing Copilot against simulated organizations without using customer data. Microsoft Copilot synthetic data addresses a difficult conflict: workplace assistants need realistic tests, while confidential business records cannot become unrestricted research material.

New tools and datasets

Nothing worth listing this week.

Worth a look (lower priority)

- Pure Differentially Private Query Release via Laplace Input Smoothing — Computer Science > Data Structures and Algorithms [Submitted on 3 Nov 2026] Title:Pure Differentially Private Query Release via Laplace Input Smoothing View PDF HTML (experimental) Abstract:We study the offline release of a public family of